Class 30

# The Fundamental Theorem of Calculus, Part 2

**Today's Goals**

- You should be able to explain why two antiderivatives of the same function $f$ differ by a constant.
- You should be able to use part 2 of the Fundamental Theorem of Calculus to calculate definite integrals.
- You should be able to find antiderivatives of basic functions.
- You should be able to distinguish between definite integrals, indefinite integrals, and antiderivatives.

1. (Blue Bell revisited) At the Blue Bell ice cream factory, milk is stored in a 40,000 gallon tank. The rate of milk flowing into and out of the tank $t$ hours after noon one day is $f(t) = 1000 - 300t^2$ gallons per hour. At 2 pm, there are 15,000 gallons of milk in the tank.

   (a) Express the net change in the amount of milk between 1 pm and 3 pm as a definite integral.

   **Solution**

   $\int_1^3 f(t)\,dt.$

   (b) Let $M(t)$ be the amount of milk in the tank at time $t$. What is $M'(t)$? What is $M(2)$? Using these two pieces of information, can you find a formula for $M(t)$?

   **Solution**

   $M'(t)$ is the (instantaneous) rate of change of the amount of milk in the tank at time $t$, which is exactly $f(t) = 1000 - 300t^2$. $M(2)$ is the amount of milk in the tank at 2 pm, which is 15000. We'd like to use the two pieces of information $M'(t) = 1000 - 300t^2$ and $M(2) = 15000$ to find $M(t)$.

   First, since $M'(t) = 1000 - 300t^2$, $M(t)$ is a function whose derivative is $1000 - 300t^2$; the only such functions are $1000t - 100t^3 + C$ for some constant $C$. We can use the fact that $M(2) = 15000$ to solve for $C$:

$$\begin{aligned} M(2) &= 15000 \\ 1000 \cdot 2 - 100 \cdot 2^3 + C &= 15000 \\ 2000 - 800 + C &= 15000 \\ C &= 15000 - 2000 + 800 = 13800 \end{aligned}$$

   So, $M(t) = \boxed{1000t - 100t^3 + 13800}$.

   (c) Use your answer to (b) to find the net change in the amount of milk in the tank between 1 pm and 3 pm.

**Solution**

Now that we have a formula for $M(t)$, we can use it to find this net change. After all, this net change simply means (amount of milk at 3 pm) − (amount of milk at 1 pm), which is

$$\begin{aligned} M(3) - M(1) &= \left(1000 \cdot 3 - 100 \cdot 3^3 + 13800\right) - \left(1000 \cdot 1 - 100 \cdot 1^3 + 13800\right) \\ &= \boxed{-600} \end{aligned}$$

**The Fundamental Theorem of Calculus, Part 2 (FTOC 2)**

If $f$ is a continuous function on $[a, b]$ and $F$ is any antiderivative of $f$ (that is, $F$ is any function such that $F' = f$), then

$$\int_a^b f(t)\,dt = \underline{F(b) - F(a) \qquad\qquad}.$$

2. Evaluate $\displaystyle\int_{-2}^{2} 4x^3\,dx$.

**Solution**

By FTOC 2, if $F$ is *any* antiderivative of $4x^3$, then $\displaystyle\int_{-1}^{2} 4x^3\,dx = F(2) - F(-2)$. So, let's try to think of an antiderivative of $4x^3$; that is, we want to think of a function whose derivative is $4x^3$. One example is $F(x) = x^4$. So, $\displaystyle\int_{-2}^{2} 4x^3\,dx = x^4\Big|_{-2}^{2} = 2^4 - (-2)^4 = \boxed{0}$.

3. Evaluate $\displaystyle\int_{4}^{3} x\,dx$.

**Solution**

One antiderivative of $x$ is $\frac{1}{2}x^2$, so $\displaystyle\int_4^3 x\,dx = \frac{1}{2}x^2\Big|_4^3 = \frac{1}{2}\cdot 3^2 - \frac{1}{2}\cdot 4^2 = \boxed{-\frac{7}{2}}$.

**Indefinite Integrals**

We use the notation $\displaystyle\int f(x)\,dx$ to mean "all antiderivatives of $f$" or "the most general antiderivative of $f(x)$" and call this the <u>indefinite integral of $f(x)$</u>. While the definite integral $\displaystyle\int_a^b f(x)\,dx$ is a number, the indefinite integral $\displaystyle\int f(x)\,dx$ is a family of functions.

4. Find each of the following.

   (a) $\displaystyle\int (4x^2 + 2)\,dx$

**Solution**

$\frac{4}{3}x^3 + 2x + C$.

(b) $\int u\sqrt{u}\, du$

**Solution**

We don't have a Product Rule for antiderivatives, so we should first rewrite this as $\int u^{3/2}\, du$; then, we can see that it's $\boxed{\frac{2}{5}u^{5/2} + C}$.

5. First decide whether each of the following definite integrals should be positive or negative; then evaluate them. Simplify your answers.

(a) $\int_1^0 (4x^2 + 2)\, dx$

**Solution**

Here, the integrand $4x^2 + 2$ is always positive, but the integral goes from 1 to 0 (so a larger number to a smaller number), which makes the integral $\boxed{\text{negative}}$. It's equal to

$$\begin{aligned} \int_1^0 (4x^2 + 2)\, dx &= \frac{4}{3}x^3 + 2x \Big|_1^0 \\ &= 0 - \left(\frac{4}{3} + 2\right) \\ &= \boxed{-\frac{10}{3}} \end{aligned}$$

(b) $\int_4^{\pi^2} u\sqrt{u}\, du$

**Solution**

$u\sqrt{u}$ is positive on $[4, \pi^2]$, so this integral is $\boxed{\text{positive}}$. It's equal to

$$\begin{aligned}\int_4^{\pi^2} u\sqrt{u}\,du &= \int_4^{\pi^2} u^{3/2}\,du \\ &= \frac{2}{5}u^{5/2}\Big|_4^{\pi^2} \\ &= \frac{2}{5}\left(\pi^2\right)^{5/2} - \frac{2}{5}\cdot 4^{5/2} \\ &= \frac{2}{5}\left(\pi^5 - \left(4^{1/2}\right)^5\right) \\ &= \boxed{\frac{2}{5}\left(\pi^5 - 32\right)}\end{aligned}$$

6. A bronze statue sits in a courtyard. One sunny day, the statue's temperature at noon is 40 degrees Celsius. Suppose that the statue's temperature is changing at a rate of $r(t) = \frac{1}{e^t} + 2$ degrees Celsius per hour, where $t$ is the numbers of hours since noon.

(a) At what times is the statue heating up? At what times is the statue cooling down?

**Solution**

$r(t)$ is always positive, so the statue is $\boxed{\text{always heating up}}$.

(b) What is the statue's temperature at 1:00 pm?

**Solution**

The statue's temperature at 1:00 pm is $40 + \int_0^1 r(t)\,dt = 40 + \int_0^1 \left(\frac{1}{e^t} + 2\right)dt$. To think of an antiderivative of $\frac{1}{e^t}$, it helps to rewrite it as $e^{-t}$; then, an antiderivative is $-e^{-t}$. SO,

$$\begin{aligned}40 + \int_0^1 \left(\frac{1}{e^t} + 2\right)dt &= 40 + \left(-e^{-t} + 2t\Big|_0^1\right) \\ &= 40 + \left[\left(-e^{-1} + 2\right) - \left(-1 + 0\right)\right] \\ &= \boxed{43 - \frac{1}{e}}\end{aligned}$$

(c) Do a consistency check: given that the statue's temperature at noon was 40 degrees, do your answers make sense with each other?

**Solution**

Yes. Since $e$ is between 2 and 3, $43 - \frac{1}{e}$ is definitely greater than 40, which is consistent with the fact that the statue is always heating up.

7. (a) Which of the following are antiderivatives of $f(x) = \dfrac{1}{\sqrt{1-x^2}}$?

$\arcsin x$ $\qquad$ $\arccos x$ $\qquad$ $\arctan x$ $\qquad$ $\arcsin x + 7$ $\qquad$ $-\arccos x$

**Solution**

We can check by taking the derivative of each: $\arcsin x$, $\arcsin x + 7$, and $-\arccos x$ all have derivative $\frac{1}{\sqrt{1-x^2}}$. $\arccos x$ has derivative $-\frac{1}{\sqrt{1-x^2}}$, while $\arctan x$ has derivative $\frac{1}{1+x^2}$.

(b) Since $f(x) = \dfrac{1}{\sqrt{1-x^2}}$ is continuous on its domain $(-1,1)$, if $F(x)$ is one antiderivative of $f(x)$, then the most general antiderivative of $f(x)$ is $F(x)+C$. Explain why this does *not* contradict your answers to (a).

**Solution**

Since $\arcsin x$ is one antiderivative of $\frac{1}{\sqrt{1-x^2}}$, the most general antiderivative of $\frac{1}{\sqrt{1-x^2}}$ is $\arcsin x + C$. The fact that $\arcsin x + 7$ is also an antiderivative of $\frac{1}{\sqrt{1-x^2}}$ clearly agrees with this, but the fact that $-\arccos x$ is an antiderivative of $\frac{1}{\sqrt{1-x^2}}$ seems to disagree with this. However, $\arcsin x + \arccos x = \frac{\pi}{2}$, so $\arcsin x - \frac{\pi}{2} = -\arccos x$. Thus, $-\arccos x$ really is $\arcsin x$ plus a constant.

8. Let $G(x) = \displaystyle\int_{-e^{2x}}^{e^{2x}} \cos(t^2)\,dt$. Find $G'(x)$.

*Hint:* This can be done without finding an antiderivative of $\cos(t^2)$.

**Solution**

Normally, the way we evaluate an integral like $\displaystyle\int_{-e^{2x}}^{e^{2x}} \cos(t^2)\,dt$ is to think of an antiderivative $F(t)$ of $\cos(t^2)$; that is, we think of a function $F$ such that $F'(t) = \cos(t^2)$. Then, FTOC 2 says that

$$G(x) = \int_{-e^{2x}}^{e^{2x}} \cos(t^2)\,dt = F\left(e^{2x}\right) - F\left(-e^{2x}\right).$$

Even though we don't have a formula for $F$, we can still differentiate this equation; using the Chain Rule, we get

$$G'(x) = F'\left(e^{2x}\right)\cdot 2e^{2x} - F'\left(-e^{2x}\right)\cdot(-2e^{2x})$$

Although we don't have a formula for $F$, we know that $F'(t) = \cos(t^2)$ (that's exactly what it means for $F(t)$ to be an antiderivative of $\cos(t^2)$), so this is equal to

$$\begin{aligned} &= \cos(e^{4x})\cdot 2e^{2x} - \cos(e^{4x})\cdot(-2e^{2x}) \\ &= \boxed{4e^{2x}\cos(e^{4x})} \end{aligned}$$

9. True or false:

(a) If $\int_0^x f(t)\,dt = \int_0^x g(t)\,dt$ for all $x$, then $f(x) = g(x)$.

**Solution**

True; If two functions are equal, then so are their derivatives. Here, the derivative of the left side is $f(x)$ by FTOC 2; this must be equal to the derivative of the right side, which is $g(x)$ by FTOC 2.

(b) If $f'(x) = g'(x)$ for all $x$, then $f(x) = g(x)$.

**Solution**

False; $f$ and $g$ could differ by a constant. A concrete counterexample is $f(x) = x^2$ and $g(x) = x^2 + 1$.

(c) If $f(x)$ is an increasing function on $(-\infty, \infty)$, then $\int_3^x f(t)\,dt$ is increasing on $(3, \infty)$.

**Solution**

False; $\int_3^x f(t)\,dt$ is increasing when its derivative (which is $f$ by FTOC 1) is *positive*. So, for example, if $f$ is a negative increasing function like $f(x) = -e^{-x}$, the statement is false.